USC Viterbi School of Engineering
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Passive Control of Aerodynamic Flows by Turbulators and Helmholtz Resonance

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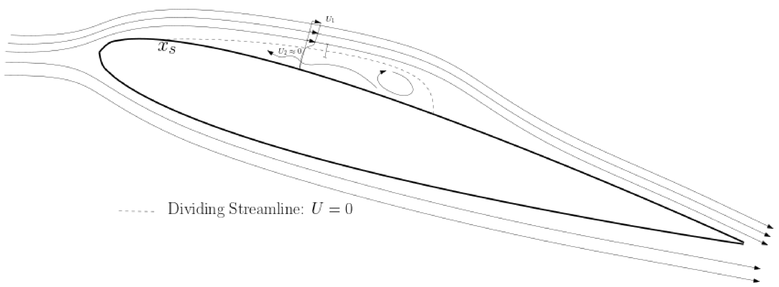
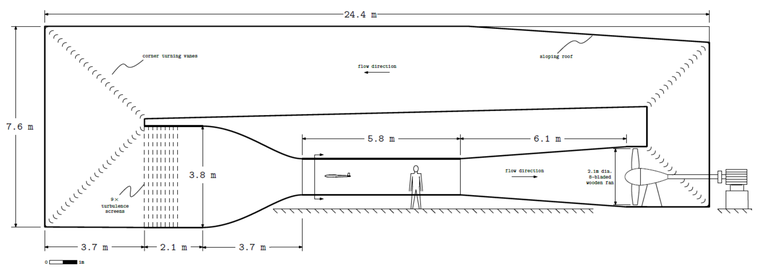
MOTIVATION & BACKGROUND

Airfoil performance in transitional Reynolds number (Re) regimes is difficult to predict due to complex boundary layer behavior, necessitating experiments for flow control development. While not completely resolving these challenges, intentional modifications to an airfoil's suction surface—using acoustic waves or boundary layer trips—offer methods to manipulate boundary layer dynamics by reducing flow separation. This study examines NACA 65-1412 airfoil aerodynamics with smooth and tripped surfaces at transitional Re. Effects of localized flow control on performance, particularly lift improvements, were assessed through baseline testing and various trip configurations. Helmholtz resonator structures were also tested as alternative local flow control methods. Tripped configurations showed significant lift improvements over smooth baseline tests.



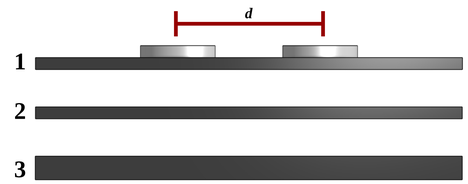
Streamline Flow over a Laminar Separation Bubble

DRYDEN WIND TUNNEL

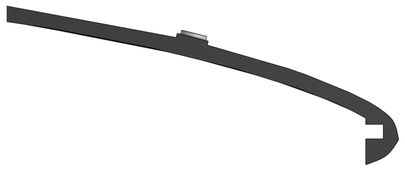


EXPERIMENTAL SETUP

Airfoil Trip Configurations

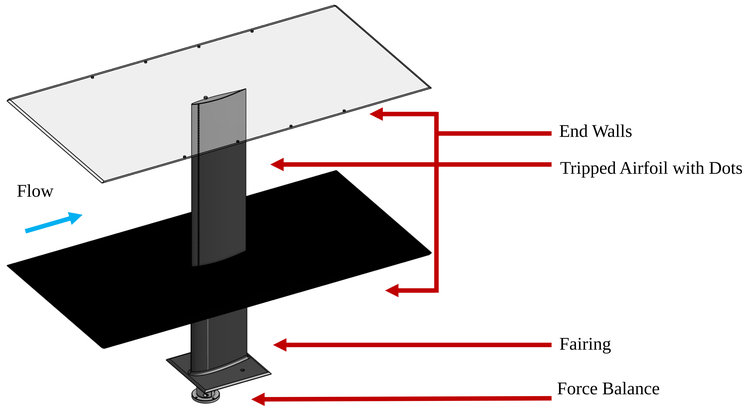


Tripped Airfoil with Dots

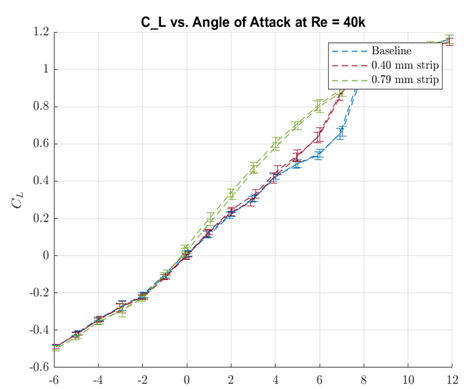


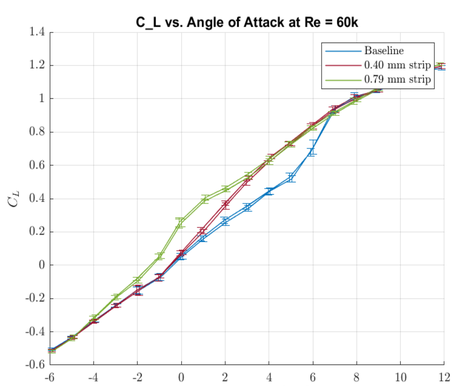
Airfoil Trip Configurations		
1	0.40mm	0.40mm Dots
2	0.40mm	No Dots
3	0.79mm	No Dots

Trip Boundary Layer Height at 0.1c	
Trip Height h (mm)	h/θ
0.79	3.8
0.40	1.9

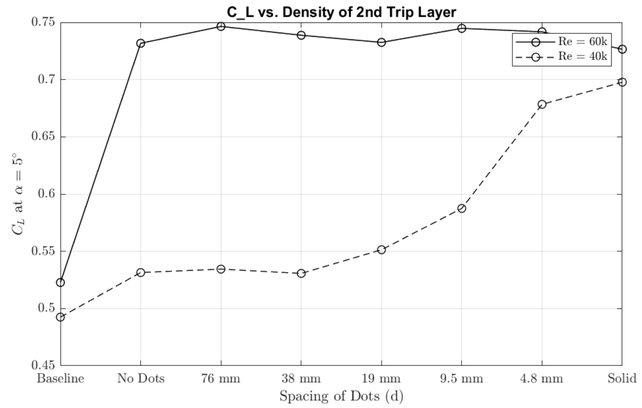


RESULTS





RESULTS



CONCLUSIONS

- Wind tunnel test results showed that more aggressive trips result in a higher C_L over wide ranges of angles of attack
- In contrast, Helmholtz resonators did not result in measurable performance changes
- At low α, Re = 60k, only the 0.79mm strip triggers full transition to the high-lift state
- At higher Re, all trip configurations trigger full transition to the high-lift state at α = 5°
- At lower Re, decreasing trip density results in a gradual decrease in C_L
- Trips only result in performance increases for angles of attack below natural laminar separation bubble formation

FUTURE WORK

- Other spanwise-non-uniform trip configurations that could further improve reattachment will be explored
- Helmholtz resonators will be tested in different configurations with a better throat-to-volume ratio
- Active forcing via embedded speakers will be compared to passive forcing methods